

Nicolas Nytko

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EDUCATION

University of Illinois at Urbana-Champaign Doctor of Philosophy, Computer Science Advised by Luke Olson and Matthew West Thesis: <i>Exploiting Hierarchy in Computational Methods:</i> Multilevel Solvers and Learning Techniques for High-Performance Computing	Aug 2020 - Dec 2026 (Expected) GPA 3.94 / 4.00
University of Illinois at Urbana-Champaign Bachelor of Science, Computer Science Grainger College of Engineering	Aug 2017 - Dec 2019 GPA 3.72 / 4.00
Harper College Transfer, Engineering Pathways	Aug 2015 - May 2017 GPA 3.59 / 4.00

PROFESSIONAL AND RESEARCH EXPERIENCE

Center for Exascale-enabled Scramjet Design <i>Graduate Research Assistant</i>	Aug 2024 - Sept 2026 <i>Urbana, IL</i>
· Developing JFNK implicit timestepping methods for nonlinear thermal radiation using discontinuous Galerkin spatial discretizations. · Implemented lazily-evaluated multigrid solver for high performance computation on GPU clusters. · Investigated use of element-level polytopic agglomeration for spatial coarsening.	
Los Alamos National Laboratory <i>Graduate Research Assistant</i>	Jun 2023 - August 2025 <i>Los Alamos, NM</i>
· Developed automatic code translation techniques for translating legacy structured PDE solvers to run on GPU accelerators, achieving 10 – 20× speed-up on key computational bottlenecks. · Developed structured auxiliary space methods for high-performance solvers on complex domains by learning diffeomorphic maps to structured domains using neural ODEs. · Researching multiresolution methods for learning problems on graph-structured data.	
Enable Medicine <i>Computational Biology Intern</i>	May 2022 - Aug 2022 <i>Menlo Park, CA</i>
· Researched numerical methods for computational biology spatial spillover correction. · Explored both data-driven and algebraic methods for correcting cell biomarker error.	
Groupon <i>Software Engineer Intern</i>	May 2021 - Aug 2021 <i>Chicago, IL</i>
Epic Systems Corporation <i>Software Developer and Intern</i>	May 2018 - Aug 2019 <i>Verona, WI</i>

PUBLICATIONS

Nytko N., MacLachlan S., Moulton J. D., Olson L. N., Reisner A., West M., *Unstructured to structured: geometric multigrid on complex domains via domain remapping*, Electronic Transactions on Numerical Analysis

Nytko N., Reisner, A., Moulton, J., Olson L., West M., *Teaching An Old Dog New Tricks: Porting Legacy Code to Heterogeneous Compute Architectures With Automated Code Translation*, International Journal of High Performance Computing Applications

Nytko N., Taghibakhshi A., Zaman T., MacLachlan S., Olson L. N., West M., *Optimized Sparse Matrix Operations for Reverse Mode Automatic Differentiation*, SIAM Journal on Scientific Computing

Zaman T., **Nytko N.**, Taghibakhshi A., MacLachlan S., Olson L., West M., *Generalizing Lloyd's algorithm for graph clustering*, SIAM Journal on Scientific Computing

Taghibakhshi A., **Nytko N.**, Zaman T., MacLachlan S., Olson L. N., West M., *MG-GNN: Multigrid Graph Neural Networks for Learning Multilevel Domain Decomposition Methods*, ICML 2023

Zaman T., **Nytko N.**, Taghibakhshi A., MacLachlan S., Olson L. N., West M., *Generalizing Reduction-Based Algebraic Multigrid*, Numerical Linear Algebra with Applications

Taghibakhshi A., **Nytko N.**, Zaman T., MacLachlan S., Olson L. N., West M., *Learning Interface Conditions in Domain Decomposition Solvers*, NeurIPS 2022

Silva M., Hieronymi P., West M., **Nytko N.**, Deshpande A., Chuang J., Hilgenfeldt S., *Innovating and modernizing a Linear Algebra class through teaching computational skills*, ASEE 2022

Silva M., Shaffer E., **Nytko N.**, Amos J., *A Case Study of Early Performance Prediction And Intervention in a Computer Science Course*, ASEE 2020

Nytko N., West M., Silva M., *A Simple and Efficient Markup Tool to Generate Drawing-based Online Assessments*, ASEE 2020

PRESENTATIONS, WORKSHOPS

41st Colorado Conference On Iterative and Multigrid Methods. *Hierarchical Graph Networks for Scalable Learning on Graphs*. June 25, 2026

22nd Copper Mountain Conference on Multigrid Methods. *Geometric Multigrid on Complex Domains via Mesh Remapping*. Apr 16, 2025

SIAM Workshop Series, UIUC. *Introduction to Machine Learning with PyTorch*. Nov 12, 2024

18th Copper Mountain Conference On Iterative Methods. *Learning sparse iterative methods through gradient-based optimization*. Apr 17, 2024

21st Copper Mountain Conference On Multigrid Methods. *Gradient-based optimization of sparse relaxation schemes*. Apr 17, 2023

17th Copper Mountain Conference On Iterative Methods. *Learning Aggregates and Interpolation for Algebraic Multigrid*. Apr 6, 2022

Hands-On With CSE, UIUC. *Building a Neural Network with PyTorch*. Apr 8, 2021

20th Copper Mountain Conference On Multigrid Methods. *A Supervised Learning Approach to Predicting Multigrid Convergence*. Mar 29, 2021

AE3 Lightning Symposium, UIUC. *Creating Drawing Questions in PrairieLearn*. Oct 3, 2019

HONORS AND AWARDS

Student paper winner: 22nd Copper Mountain Conference on Multigrid Methods April 2025

Andrew & Shana Laursen Fellowship Fall 2022 - Spring 2023

Outstanding Teaching Assistant	Spring 2021
Outstanding Course Assistant	Fall 2018
Dean's List, College of Engineering	Fall 2017, Spring 2018, Spring 2019

TEACHING EXPERIENCE

Graduate Teaching (UIUC)

CS 450 - Numerical Analysis (Fall 2023 - Spring 2024)

CS 357 - Numerical Methods I (Fall 2020 - Spring 2022)

MATH 257 - Linear Algebra (Computational Section Pilot) (Spring 2020)

CS 199 - Python for Data (Fall 2019 - Spring 2020)

Undergraduate Teaching (UIUC)

CS 357 - Numerical Methods I (Fall 2018 - Fall 2019)

CS 126 - Software Design Studio (Spring 2018)

ACTIVITIES

President, SIAM UIUC chapter	Fall 2024 - Spring 2026
President, Harper Society of Engineers	Spring 2017